



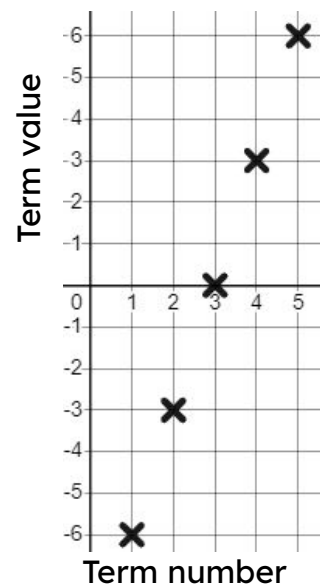
Representing sequences graphically

Task A: Plotting arithmetic sequences

1) Find and plot the first five terms of these arithmetic sequences.

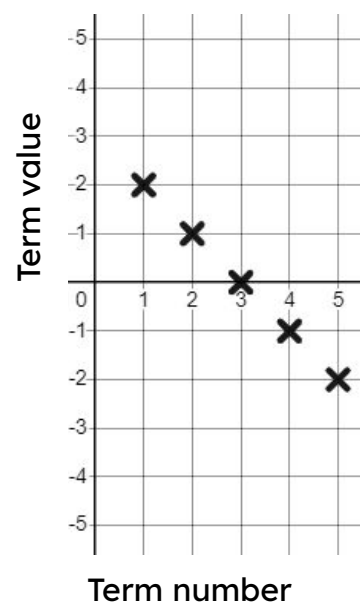
a) $3n - 9$

Term number (n)	1	2	3	4	5
Term value (T)	-6	-3	0	3	6



b) $3 - n$

Term number (n)	1	2	3	4	5
Term value (T)	2	1	0	-1	-2



2) Match the sequences to their graph.

$4n + 10$
Increasing
sequence
with all
positive
term values.

$4n - 10$
Increasing
sequence
with some
negative
term values.

$10 - 4n$
Decreasing
sequence.



Task B: Features of sequences represented graphically

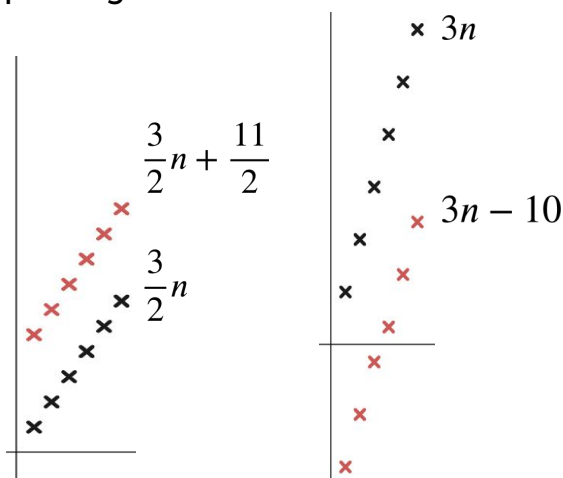
1) a) Match these arithmetic sequences to the correct plotting.

b) Write a few sentences explaining how you identified each sequence.

You might have written:

The sequences $3n$ and $3n - 10$ have the same increases term-to-term, with $3n - 10$ being a translation of $3n$ by 10 in the negative vertical direction.

$\frac{3}{2}n + \frac{11}{2}$ has the same term-to-term increase as $\frac{3}{2}n$ but, is a translation by $\frac{11}{2}$ in the positive vertical direction.



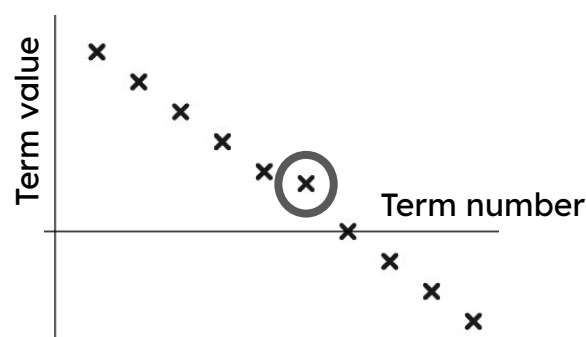
2) Izzy has plotted the sequence $5 - \frac{5}{7}n$

Which term number is wrong? Write a sentence to justify your answer.



The 6th term number is wrong.

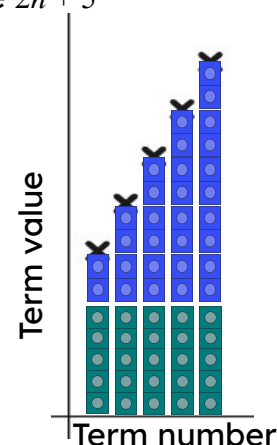
$5 - \frac{5}{7}n$ is an arithmetic sequence. The points should align and the 6th one is not in line.



3) a) Sofia has started to draw the structure of the arithmetic sequence $2n + 5$. Finish her diagram for her.

b) Write a sentence explaining why this structure represents the sequence $2n + 5$.

You might have written, the 5 remains constant whereas $2n$ is a variable, varying the number of 2's.



4) The 10th term of the arithmetic sequence $3.5n$ is 35 and the 20th term is 70 so Laura says...



I know the 10th term of the arithmetic sequence $3.5n - 21$ is 14 so the 20th term must be double that. It's 28

Explain to Laura why this does work for the arithmetic sequence $3.5n$ but it doesn't for the sequence $3.5n - 21$.

You might have said, the sequence $3.5n$ has no constant so it is just twice as many variables, whereas $3.5n - 21$ contains a constant and that constant will not double.

